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Deep Neural Networks Methods applied to Face-based Emotion Recognition

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June 18, 2024

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Abstract

The task of recognizing an emotion through someone's facial expressions is natural for humans as we perform that task on a daily basis. On the other hand, this has not been yet an easy job for computers in terms of human-computer interactions. Upcoming research delves into the development and study of better models and datasets in order to reach a point where this recognition is as effective for machines as it is for humans.
This report aims to explore the use of a landmarked representation of the image samples of a crowd-sourced label dataset that is FER+, on a deep learning neural network that is not very deep but present good results on the task of face emotion recognition, the ResNet-18. The goal is to recognize if a simpler representation of a facial expression could still be enough for its classification, factoring in the impact of the dataset's class distribution. Furthermore, an evaluation of two different facial features, eyes and mouth, allowed to infer about their contributions to the representation of emotions.
The conclusions obtained indicate a clear drop on the performance of the model for the landmarked images in relation to the original image samples. However, as we appealed to the representation of a facial expression by the landmarks of the major facial features, the performance remained almost the same, indicating a clear statement that those facial features are enough for emotion classification through facial expressions and can be used instead of all 468 landmarks. In terms of the performance with only eye or mouth landmarks, there was a clear difference, whether in the accuracy of the predictions or which emotions each feature better represents. Ultimately, it was concluded that the relationship between the facial features is highly important in the recognition of facial expression.

Resumo

A tarefa de reconhecer uma emoção através das expressões faciais é natural para os humanos, já que realizamos essa tarefa diariamente. Por outro lado, isso ainda não tem sido uma tarefa fácil para os computadores em termos de interações humano-computador. Pesquisas futuras exploram o desenvolvimento e estudo de modelos e conjuntos de dados melhores para alcançar um ponto em que esse reconhecimento seja tão eficaz para as máquinas quanto é para os humanos.

Este relatório tem como objetivo explorar o uso de uma representação com pontos fiduciais das imagens de um conjunto de dados, etiquetados por vários anotadores, denominado FER+, numa rede neural de *deep learning* não muito profunda, mas que apresenta bons resultados na tarefa de reconhecimento de emoções faciais, a *ResNet-18*. O intuito é reconhecer se uma representação mais simples de uma expressão facial ainda seria suficiente para sua classificação, levando em conta o impacto da distribuição de classes do conjunto de dados. Além disso, uma avaliação de duas características faciais diferentes, olhos e boca, permitiu inferir sobre suas contribuições para a representação de emoções.

As conclusões obtidas indicam uma clara queda no desempenho do modelo para as imagens com pontos fiduciais em relação às amostras de imagem originais. No entanto, ao recorrermos à representação de uma expressão facial pelos pontos das principais características faciais, o desempenho permaneceu quase o mesmo, indicando uma afirmação clara de que essas características faciais são suficientes para a classificação de emoções através das expressões faciais e podem ser usadas em vez de todos os 468 pontos. Em termos de desempenho com apenas pontos fiduciais de olhos ou boca, houve uma clara diferença, tanto na precisão das previsões quanto em quais emoções cada característica representa melhor. Em última análise, foi concluído que a relação entre as características faciais é altamente importante no reconhecimento da expressão facial.

Acknowledgements

On a first instance, I would like to recognize the assistance given by Ana Filipa Sequeira and Pedro Neto throughout the semester, by helping me with my work and clarifying any of my doubts.
I would also like to thank CTM - Centro de Telecomunicações e Multimédia, and in particular the VCMI (Visual Computing and Machine Intelligence) group from INESC TEC, for introducing me to the topic of biometrics and all the support given.

Leandro Ribeiro

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Abreviaturas e Símbolos

Acc	Accuracy
AFEW	Acted Facial Expression in the Wild
AFER	Automated Face Expression Recognition
ASD	Autism Spectrum Disorder
ASM	Active Shape Mode
CEL	Cross-Entropy Loss
CK+	Extended Cohn-Kanade
CNN	Convolutional Neural Network
DBN	Dynamic Bayesian Network
DL	Deep Learning
FER	Face Emotion Recognition
GANs	Generative Adversarial Networks
HCI	Human-Computer Interaction
HOG	Histogram of Oriented Gradients
JAFFE	Japanese Female Facial Expression
LBP	Local Binary Patterns
ML	Multi-label Learning
MLP	Multi-Layer Perceptrons
MV	Majority Voting
NF	Not Found
NNB	Neural Network Based
PAD	Pleasure-Arousal-Dominance
PCA	Principal Component Analysis
PLD	Probabilistic Label Drawing
RAFDB	Real-world Affective Faces Database
ResNet	Residual Neural Network
RF	Random Forest
SFEW	Static Facial Expression in the Wild
SGD	Stochastic Gradient Descent
SSD	Single Shot Detector
SVM	Support Vector Machine
VA	Valence-Arousal

Chapter 1

Introduction

1.1 Context

Emotions are crucial to every aspect of human life, influencing our decisions, behaviors, and interactions. They can be defined as a conscious mental reaction (such as anger or fear) subjectively experienced as strong feeling usually directed toward a specific object and typically accompanied by physiological and behavioral changes in the body [8].

Despite the fact that humans are more than accustomed with emotion recognition on a daily basis, the definition of basic emotions was only done in 1972 by Ekman and Friesen based on a cross-cultural study, that stated that there were 6 basic emotions that are expressed and interpreted the same way across all cultures: happiness, anger, fear, disgust, surprise and sadness. Later on, contempt was added to the list of basic emotions. In 1990, emotions such as shame, guilt, excitement and even arrogance were included in this record. The previously stated basic emotions can be aggregated in what are called compound/nonbasic emotions such as happily surprised or sadly angry. This division is included in a discrete representation model. In a dimensional representation model, emotions are defined as points in the dimensional space. The Valence-Arousal(VA) and Pleasure-Arousal-Dominance (PAD) are two typical and widely accepted dimensional emotion models [9].

Some primary methods for monitoring emotions are facial expressions, physiological data, speech recognition, information on personal preferences, and questionnaires, but our face is the most expressive part of our body and is popularly known as an index of our mind due to its highly expressive way of demonstrating emotions. Therefore, facial expression serve as a social communicator and plays a major role in nonverbal communication [10].

According to research, the verbal component of a message, or uttered words, contributes 7%, the vocal part contributes 38%, and the speaker's facial expressions contribute 55% to the overall effect of the spoken message [10]. As a result, researchers directed their focus to the study of face emotion recognition assisted by computer vision methodologies.

Affective computing is a field of computer science that aims to study and develop the theories, methods and systems that can detect, understand, and replicate human emotions. It is one of the

most important research fields in modern human-computer interaction (HCI) and with it arises the branch of AFER (Automated Face Expression Recognition), with the purpose of automatically recognizing human emotions based on facial biometric markers [9].
The importance of Face Emotion Recognition spreads across a wide range of applications, either healthcare in the detection of diseases or in terms of suicide prevention, employment, education and even public safety, crime detection or marketing purposes. However, several factors must be taken into account, such as data protection, accuracy, fairness and so many others [11].

1.2 Motivation

Human-computer interaction has become a more researched topic over the years due to not only its ability of replacing current methodologies but also in the innovative approaches it brings. Moreover, AFER can have a huge impact in certain fields such as healthcare, related to patients inability of verbal communication, which includes people with Autism Spectrum Disorder (ASD), mother-infant interactions and also social cognition.
Despite the recurring discoveries of new methods to manner the face emotion recognition task, there is still room for improvement, whether by reducing the computational cost of certain algorithms or by increasing their accuracy, more precisely in deep-learning based strategies as it is a promising area regarding AFER.

1.3 Objectives

The primary purpose for this report is to evaluate the significance of the different facial cues in the task of face emotion recognition through the use of fiducial points, also known as facial landmarks. By assessing how well a deep learning model performs on merely facial landmarks, one can conclude if facial expressions can be represented by said points and still deliver an accurate prediction.
Secondly, a motivation to dive into Sustainable Development Goals, established by the United Nations, integrating sustainability principles to Engineering practices, in order to evaluate the ability of this project to benefit a more sustainable future, as presented in Table 1.1.

Table 1.1: SDG goals [7].

SDG	Targets	Contributions
3	4	Reduce suicide mortality rate
16	1	Lie detectors, predictive screening to identify potential terrorism threats
	5	Crime detection, spotting shoplifters, and fraud prevention strategies

Chapter 2

Fundamental Concepts

2.1 Emotion Recognition

Emotion Recognition is the process of identifying human emotion based on their facial, vocal and postural expressions and its accuracy varies from person to person. According to Scherer's Component Process Model [12], there are five crucial elements of emotion: cognitive appraisal, an evaluation of events and objects, bodily symptoms, action tendencies, expression, such as facial, vocal or gestures, and feelings, the subjective experience of emotional state. Once all these processes have been coordinated and synchronized, an emotional experience has occurred. Finding all the necessary physical cues to recognize an emotion might be difficult but an adequate observation may be found by looking at the facial expression. As stated before, facial expressions contribute to 55% for the overall effect of the spoken message [13].

Figure 2.1: Happiness and Sadness from FER+ Database [1]

Facial expressions is a term regarding the recurring configurations of facial muscle movements that communicate some thought, emotion or behaviour, for instance, a person's eyebrows raised in the middle may be acknowledged as sympathy or flashing both eyebrows upward be recognized as a greeting. Regarding emotions, by raising lip corners, one may be showing happiness, or sadness by frowning.
The systematic study of facial expression started with Charles Darwin when he published *The Expression of the Emotions in Man and Animals* [14] where he proposed that humans across all cultures have particular and distinct facial expressions for particular emotions, which are produced involuntarily as behavioral and physiological reactions. Following evidence supported this notion as two researchers, Plutchik and Tomkins [15], displayed photographs of people posing prototypical emotions and observers would agree as to which expression represented which emotion.

Finally, observation of children born blind and deaf favored universality as they showed similar expressions of emotion as their counterparts. In the end, Ekman [16] argued that basic human emotions generated particular patterns of physiology and facial expressions which were universal across all cultures, but that their ultimate expression was modified, exacerbated, suppressed, or masked by social learning processes dependent upon personal, family, or cultural display rules.

2.1.1 Databases

The ability of a machine learning algorithm to fulfill its purpose relies heavily on the data with which it was trained. Those similar samples are organized in what is denominated as databases, where its variability and size are important factors that affect the overall accuracy of a model and can help it have a better generalization, as having enough labeled training data that comprises as many different people as possible and under different environments is critical to training an efficient FER system.

FER databases can be static-based, dynamic-based or mixed depending on the type of input data. In a static-based approach, the features representation relies on the spatial information related to each image, where as in dynamic-based approaches, the temporal relationship among successive images is taken into account.

Static-based databases have been used more often over the years, with a special emphasis on RAFDB [17], AffectNet [18], JAFFE [19], SFEW [20] and FER2013 [21]. Real-world Affective Faces Database (RAFDB) is a large scale database with over thirty thousand highly diverse images in gender, ethnicity, age, lighting conditions and occlusions, downloaded from the Internet, where each of them has been independently labeled by forty annotators, what is called crowd-sourcing annotation. AffectNet contain over more than 1M facial images, where half of them were manually annotated for the presence of seven basic facial expressions and the intensity of valence and arousal in the wild. JAFFE corresponds to a more specific database as the subjects are only Japanese females and all were obtained on a controlled environment. Static Facial Expression in the Wild (SFEW) dataset was created by selecting static frames from the AFEW database by computing key frames based on facial point clustering. FER2013 comprises more than thirty-five thousand gray scale and greatly diversified samples. Despite that variability, this database lacks a balance between its emotional classes, as some classes such as disgust have less samples when compared to happiness.

In terms of dynamic-based databases, one that stands out is Acted Facial Expression in the Wild (AFEW), composed of different videoclips from movies, with a high variability on lighting, head poses, occlusion and spontaneous expressions.

Regarding Mixed databases, Extended Cohn-Kanade (CK+) is the most appealed database as it contains videos of individuals ranging from 18 to 50 years of age with a variety of genders and heritage, where each video shows a facial shift from the neutral expression to a targeted peak expression [10]. Despite the high number of images in the available databases and the fact that a few of them were taken in a controlled environment, those images require some preprocessing so they can be used for the training of algorithms.

2.1.2 Data Preprocessing

Raw data is not, most of the time, suitable to be directly used as input for a model. Therefore arises the need for its processing, as it improves the quality of input data, enhance the robustness of models, and contributes to a better recognition accuracy. Some of data preprocessing techniques include face detection, face alignment and face normalization.

- **Face Detection**: process of determining whether and where faces are present in a captured image. As the background of raw data includes information that has no interest for the task of face emotion recognition, it can be excluded through the use of algorithms for face detection associated with cropping the image in the area around the detected face. By discarding this useless information, there can be an increase in accuracy as the model can focus only on the features of the region of interest.
- **Face Alignment**: the process of rotating, translating and scaling a face image so that the transformed image has certain dimensions and the eyes, nose and mouth corners are approximately located at pre-defined locations in the transformed image. This standardization helps improve the accuracy and robustness of machine learning models by reducing variability caused by different head poses, scales, and rotations.
- **Face Normalization**: involves standardizing the appearance of face images to reduce illumination variation, head pose, and other influences on facial expression recognition. The goal is to align all faces to a common frame of reference, which simplifies the learning task. This can be achieved through histogram equalization, gamma intensity correction and homomorphic filtering. Normalization of image colour values is a must for CNNs with respect to the mean and standard deviation for all the colour channels [9] [22].

2.2 Deep Learning

Deep learning is a subset of machine learning that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain through a combination of data inputs, weights, and bias. They are composed of three or more layers and are trained on large amounts of data to identify and classify phenomena, recognize patterns and relationships, evaluate possibilities, and make predictions and decisions. Each layer in a deep neural network helps refine and optimize those outcomes for greater accuracy.
All the layers are interconnected and they use the output of the previous layer as its input, in order to generate an output at the end of a network, in what is regarded as forward propagation. In order to train a model, a process called backpropagation uses algorithms to calculate errors in predictions and adjust the weights and biases by moving backwards through the layers. With the help of forward and backpropagation, a neural network becomes gradually more accurate [23].
The most basic unit of a neural network is the Perceptron. It takes a set o values x_i as inputs and weights w_i, and performs their scalar product, adding a bias b. Finally, an activation function θ is

applied to decide whether a neuron should be activated or not, and the output passes to the next layer of the network.

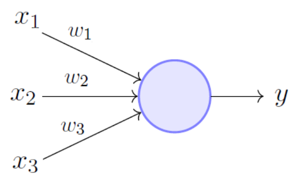


Figure 2.2: Perceptron [2]

$$f(x) = \theta\left(\sum_{i=1}^{m} w_i \cdot x_i + b\right) \tag{2.1}$$

A perceptron can also be viewed as a single-layer perceptron, the most simples feedforward neural network. Itself alone cannot perform the complicated task the human brain is capable of, so they are organized in what is called a convolutional neural network (CNN).

2.2.1 Convolutional Neural Networks

Neural networks are a subset of machine learning, comprised of layers of nodes, which include an input layers one or more hidden layers and an output layer. Each node is connected to another and has an associated weight and threshold. As stated before, if the output of a node is above a certain threshold, that node is activated and the output passes along to the next layer. Convolutional neural networks distinguish from other neural networks due to their higher performance with image, speech or audio signal inputs. They are designed to process data in the form of multiple arrays, such as colored images which have three 2D arrays with the intensity of its pixels in the three color channels. For that purpose, its architecture comprises convolutional layers and pooling layers.

- **Convolutional Layers**: It is the core building of a CNN and where the majority of computations occur. In a process called convolution, a filter/kernel, a two-dimensional array of weights, which may vary in size, also known as a feature detector, will move across an image and check if a certain feature is present, giving rise to a feature map. The scalar product is performed between the kernel weights and the inputs pixels and fed into an output array. The kernel slides until it has moved across the entire input image.

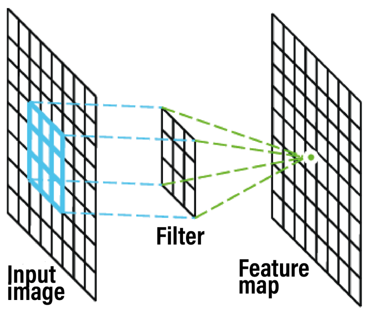


Figure 2.3: Convolution operation [3]

- **Pooling Layers**: Their role is reduce dimensionality by decreasing the number of parameter in the input. Similar to a convolutional layer, a kernel shifts through the input but in this case it applies an aggregation function to the input feature map. Max pooling is when, of all the pixels encompassed by the filter, the highest value is passed along as the output. In average pooling, the output is the average value of those pixels. Although information is lost in pooling layers, the improved efficiency, reduced risk of overfitting and reduced complexity are factors that outweight that loss.

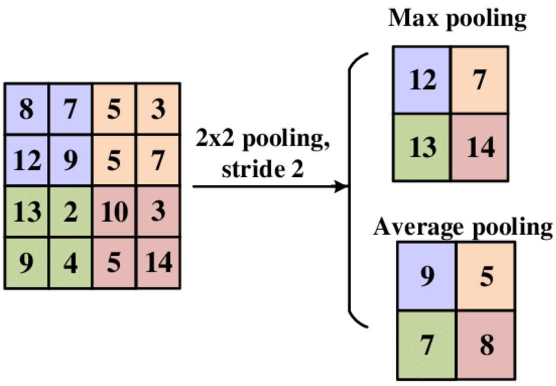


Figure 2.4: Example of Max Pooling and Average Pooling operation [4]

- **Fully Connected Layers**: In this case, all the inputs from one layer are connected to every activation unit of the next layer, as each neuron/perceptron applies a linear transformation to the input. This dense layer is responsible for the task of classification based on the previously extracted features through a softmax activation function, producing a final prediction consisting on a probability, from 0 to 1, for each class [23].

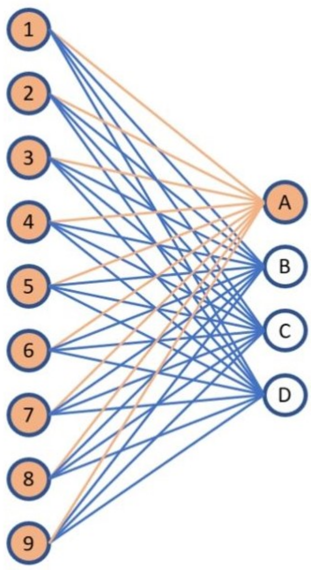


Figure 2.5: Fully Connected Layer [5]

2.2.2 Transfer Learning

Instead of training a model from scratch, which requires a large amount of data and computational power, transfer learning leverages the knowledge gained from a model that has already been trained on a large dataset by transferring the learned weights. The general idea is to use the knowledge a model has learned from a task with a vast labeled training data into a new task associated with a small set of data. This idea can significantly reduce the training time and improve performance, especially when the target dataset is small [24] [6].

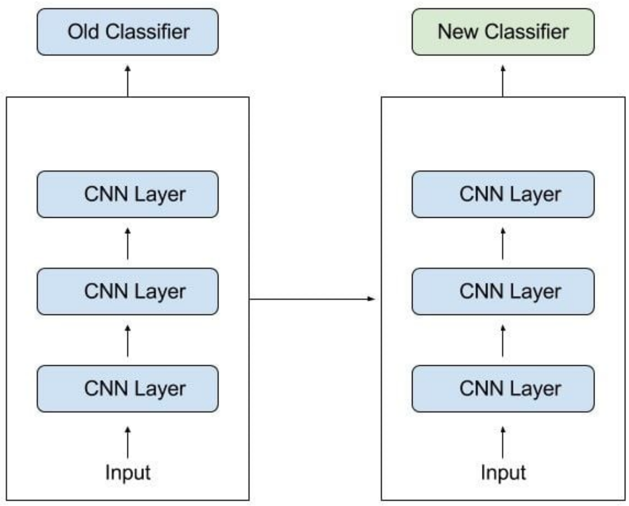


Figure 2.6: Transfer Learning [6]

2.2.3 Supervised Learning

Supervised learning is a subcategory of machine learning where labeled datasets are used to train algorithms that classify data or predict outcomes accurately with the expectation that the models learns the relationship between the inputs and outputs. A training set consisted on inputs and correct outputs, is used to teach the model to yield the desired output. A loss function helps the model adjust its parameters so the error is minimized, making an improvement on the model's accuracy. The goal is for the trained model to be capable of correctly predict an output when an unknown input is presented [25].

Chapter 3

State-of-the-art

The approach for state-of-the-art face emotion recognition follows a certain methodology with 3 fundamental steps: Face Detection, Feature Extraction and Emotion Classification.

3.1 Face Detection

Face Detection consists on determining whether and where faces are present in an image. The result is a bounding box containing the central region of interest of the face, a rectangular approximation of the landmarked region, thus being used for the estimation of landmarks [22]. It may be accompanied by face alignment, normalization, or even grayscale conversion, to rotate and frontal the detected faces, eliminate illumination variation, head pose and other factors that may influence Face Emotion Recognition, and decrease the computational cost while preserving relevant features and having no loss in terms of precision [9] [26].
One of the most used algorithms for Face Detection was developed by Paul Viola and Michael Jones. It consists of a boosted cascade of simple features that allows background regions of the image to be quickly discarded while spending more computation on promising object-like regions [27].
Deep Learning has made its way as a state-of-the-art approach for face detection with Convolutional Neural Networks (CNN) such as the Multi-task Cascade CNN architecture, where an image is viewed as a matrix of pixels that will undergo a series of convolutional filters, pooling layers and functions that will classify the image with a probabilistic value taking into account the available classes [26].

3.2 Feature Extraction

Feature Extraction aims to learn and extract the most important and descriptive piece of emotional information from images, more precisely facial expression features for emotion classification. The approaches employed for feature extraction can be partitioned into shallow learning-based and deep learning-based methods.

Shallow learning-based or handcrafted facial expression feature are associated with more tradition approaches and can be separated into geometrical and appearance features.
Geometrical features rely on the shape, size, position and direction of facial organs such as eyebrows, nose, eyes, mouth and so on. They refer to the track of facial landmarks, a set of anthropometric points in the image marking the contour and different parts of the face [22], and can be presented as fiducial points or a face mesh. The main purpose is to extract facial features by using the geometric relationship between these landmarks, ignoring other important characteristics such as skin texture change [28]. As for appearance-based methods, those rely in changes in facial texture, which captures the intensity changes associated with different expressions, such as wrinkles, bulges and furrows.
Some state-of-the-art techniques comprise Active Shape Model (ASM) [29], Gabor Filters [28], Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), achieving greater results when compared to more traditional methods such as Principal Component Analysis (PCA) [9].
Deep learning-based methods have been widely used for deep facial feature extraction because of their strong feature learning ability, by extracting high-level emotional information from images, and have become state-of-the-art approaches. CNN is often used to extract local facial features by using a set of filters to identify important patterns or features, and its performance is highly dependent on the dataset. To acquire high-level discriminative features, the training samples must be abundant and diverse. A way to bypass this issue relies in transfer learning, by fine-tuning parameters in a pre-trained deep CNN model, which was trained on a large-scale dataset. Apart from CNNs, Generative Adversarial Networks (GANs) [9] have been used in facial expression recognition for data augmentation and reducing the negative influences of emotion-related variables. Attention mechanisms [9] have been embedded in DL models to weight features according to their significance in facial expression recognition task.

3.3 Classification

The last step for face emotion recognition is determined by the classifier, which takes into account the extracted features and produces a prediction of the corresponding emotion. Some classical approaches include Support Vector Machine (SVM), the most used classifier for emotion recognition, Random Forest (RF) and Dynamic Bayesian Network (DBN) [26]. As for Neural Network Based approaches (NNB), those encompass CNN arising methods and Multi-Layer Perceptrons (MLP) [9].

Chapter 4

Methodology

The general idea behind this methodology is to obtain landmarked images from a crowd-sourced labeled dataset and train a deep learning model with said images and the original samples, and comparing the obtained metrics to evaluate the model's ability to correctly predict an emotion based solely on the landmarked images and their corresponding label. The general approach is described in 4.1.

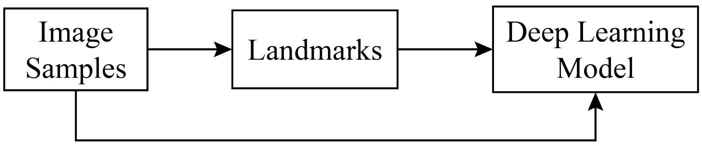


Figure 4.1: General process

For that purpose, this section is divided in two subsections, regarding the dataset preparation and the model training. The first comprises a description of the dataset used, an explanation on the selection of the images' labels, necessary when manipulating a crowd-sourced labeled dataset, and how the landmarks were obtained. The later will describe the architecture of the deep learning model used, its training procedure and finally the evaluation metrics used.

4.1 Dataset Preparation

In this work, the chosen dataset was FER+ [1], were a set of new labels for the standard Emotion FER dataset [30] was constructed with the contribution of ten taggers. The original FER data set was prepared by Pierre Luc Carrier and Aaron Courville by searching on the Internet for face images with emotion related keywords. The images were filtered by human labelers, but the label accuracy is not very high. That said, an approach regarding crowd-sourced labeling was considered, in the hopes of obtaining better quality ground truth for still image emotion. Having 10 taggers for each image enables researchers to estimate an emotion probability distribution per face. This allows the construction of algorithms that produce statistical distributions or multi-label outputs instead of the conventional single-label output.

Figure 4.2: Examples of FER+ images' classes [1]

The images of size 48x48x3 are divided in three sets, train, validation and test, each with 28558, 3579 and 3573 images respectively, and each tagger was required to choose an emotion for each one, Neutral, Happiness, Surprise, Sadness, Anger, Disgust, Fear or Contempt. In cases of uncertainty, there was the choice of Unknown, and if there was no face available, NF (not found) [31] [1].

4.1.1 Labels

As stated before, each image in the FER+ dataset was labeled by ten annotators. Factoring in the input of our deep learning model, an image and its label, emerges the need to, between the ten labels indicated by the taggers, choose only one. Using this same dataset, Barsoum (2016) explored 4 different approaches, majority voting (MV), multi-label learning (ML), probabilistic label drawing (PLD) and cross-entropy loss (CEL) and his conclusions indicate that MV is a reliable option.
Let there be a total of N training examples I_i; i = 1, ... , N. For the i^{th} example, let the network's output after its soft-max layer be $q^i{}_{\text{k}}$, k = 1, ..., 8, and the crowd-sourced label distribution for this example be $p^i{}_{\text{k}}$, k = 1, ..., 8. Naturally, we have:

$$\sum_{k=1}^{8} q^i{}_{\text{k}} = 1; \qquad \sum_{k=1}^{8} p^i{}_{\text{k}} = 1. \tag{4.1}$$

In MV, the goal is to arrive at a final label based on the majority opinion or consensus among the annotations. A new target distribution $\hat{p}^i{}_{\text{k}}$ for each I_i such that:

$$\hat{p}^i{}_{\text{k}} = \begin{cases} 1 & \text{if } k = \text{arg max}_{\text{j}} p^i{}_{\text{j}} \\ 0 & \mathit{otherwise} \end{cases}$$

That said, each image was labeled according to the most voted class between the taggers. In the few instances where there is a tie, the chosen class is the first in the draw, according to the sequence Neutral, Happiness, Surprise, Sadness, Anger, Disgust, Fear, Contempt, Unknown or NF [1]. At the end, an analysis of the dataset's class distribution was performed.

4.1.2 Landmarks

The acquisition of the facial landmarks was carried out using Google's MediaPipe Solutions [32], a library that offers a collection of solutions for multimedia processing such as audio or video, with a focus on computer vision and machine learning.
The first step is done by a face detection machine learning model, the BlazeFace short-range model, a lightweight and accurate face detector, optimized for front-facing images at short range. Its architecture uses a Single Shot Detector (SSD) convolutional network technique with a custom encoder [33]. The model outputs face locations and the information for the bounding box, along with the following facial key points: left eye, right eye, nose tip, mouth, left eye tragion, and right eye tragion. If a face is detected, then a face mesh model is responsible for creating a face mesh, a detailed 3D representation of a human face, and outputting the 468 3-dimensional face landmarks, whose coordinates x and y are normalized and range from 0 to 1 [33]. The final step includes rescaling and drawing the landmarks in a previously created black image that serves as the background.

The drawn face landmarks are dependent on what facial elements are intended to be used for emotion recognition. That said, four different "datasets" with landmarked images were created: one with all the 468 face landmarks obtained from MediaPipe, a second one with the landmarks for the eyes, eyebrows, nose and mouth, the third with only landmarks from the eyes and the last with landmarks from the mouth.

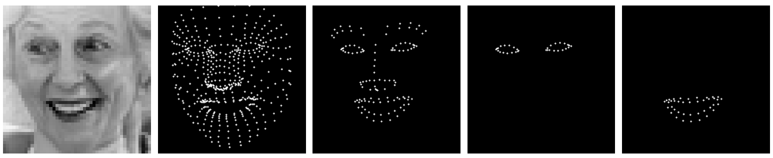


Figure 4.3: Examples of the original image [1] and the corresponding 468 landmarks, eyes, eyebrows, nose and mouth's landmarks, eyes' landmarks and mouth's landmarks

4.2 Model Training

4.2.1 Model Architecture - ResNet-18

The ResNet (Residual Neural Network) architecture was introduced by Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun in their paper titled "Deep residual Learning for Image Recognition " in 2015 [34]. This new architecture for convolutional neural networks excels in the task of image classification by addressing some issues when training very deep neural networks. One of its important features is the residual block. In a classical neural network, the input is transformed by a set of convolutional layers then it is passed to the activation function. In a residual network, the input to the block is added to the output of the block by a skip connection, allowing the gradient of the loss function to flow more easily. ResNet architectures are formed by stacking multiple of these residual blocks together, creating very deep neural networks, and typically utilise global average pooling as the final layer before the fully connected layer.

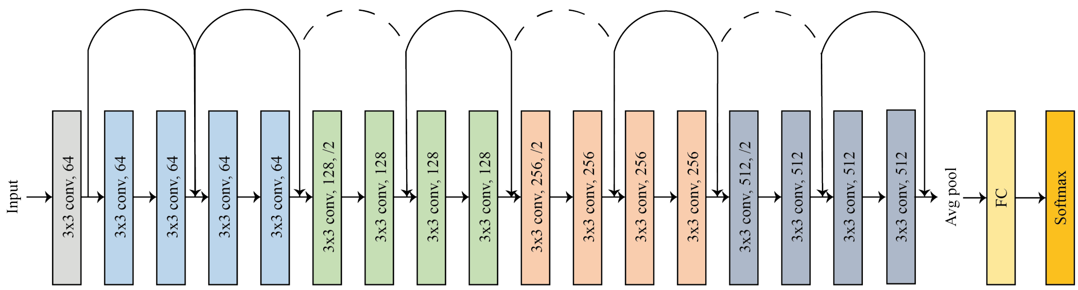


Figure 4.4: ResNet-18 architecture [4]

As the name states, the chosen deep learning model for this work, ResNet-18 has 18 layers, including convolutional layers, batch normalization, and ReLU activation functions. It is a pretrained network on more than one million images from the ImageNet database and takes as input an RGB image of size 224x224x3 [35] [36].

4.2.2 Training Procedure

Before the training process starts, each image must be resized to the dimensions 224x224x3 so they can be used as input for the model, followed by a normalization of the image colour values according to Table 4.1. After that, the data loader, a component that handles the loading and preparation of data, is built with shuffling and a batch size of 32. This means that the input of our network will be batches of 32 sequential images at a time where its order will change through every to ensure and efficient management of memory and that the network does not rely its predictions on the order of the images rather then their features.

Table 4.1: Normalization values for ResNet-18

Parameter	Channel 1	Channel 2	Channel 3
Mean	0.485	0.456	0.406
Standard deviation	0.229	0.224	0.225

The next step consists on loading the pretrained network and changing the number of classes of the fully connected layer, responsible for the classification, to the number of classes we are using, which means, going from 1000 to only 9, Neutral, Happiness, Surprise, Sadness, Anger, Disgust, Fear, Contempt and Unknown. Subsequently it is necessary to define the loss function and the optimizer. The loss is responsible for measuring the difference between the predicted output of a model and the actual target value. The optimizer updates the weights of the network in the attempt to minimize the loss function and therefore enhancing the performance. The chosen loss function was a standard Cross-Entropy Loss described by equation 4.2, where $\hat{p}^{i}{}_{k}$ corresponds to the target distribution and $q^{i}{}_{k}$ the soft-max probability for the i^{th} class. As for the optimizer, it implements the Stochastic Gradient Descent, that updates the model parameters in the direction of the negative gradient of the loss function, with a learning rate of 0.001, that sets the size of the steps the optimizer takes during each update, and a momentum of 0.9, that helps accelerate convergence.

$$\mathscr{L} = -\sum_{i=1}^{N}\sum_{k=1}^{8} \hat{p}^{i}{}_{k} \log q^{i}{}_{k}; \tag{4.2}$$

The training of the ResNet-18 neural network has a duration of 20 or 30 epochs depending on the "datasets" used, the latter for the images with only the landmarks of the mouth or the eyes. The rest of the sets were trained with 20 epochs. This means that the training and validation data performed 20 or 30 full passes through the network's architecture and the weights are updated at each training phase but not on validation phases, as its purpose is to evaluate the model's performance on a separate subset of data to help in assessing how well it generalizes to unseen data and aid in fine-tuning its parameters and hyperparameters.

4.2.3 Evaluation Metrics

After each training and validation phases, the accuracy and balanced accuracy are calculated for each of them. At the end, a test phase was performed to evaluate the final performance of the network, with the calculation of its accuracy and balanced accuracy.
The accuracy will indicate the performance of the network by evaluating the amount of correctly predicted instances out of the total instances in the dataset.

$$Acc = \frac{Number\ of\ true\ predictions}{Total\ number\ of\ predictions} \tag{4.3}$$

As for the balanced accuracy, it is used to evaluate the performance but taking into account the imbalance of the class distribution of the dataset, resulting in the average of the recall obtained on each class. Take C as the total number of classes, TP_i as the number of correct predictions of the i_{th} class and FN_i as the number of incorrect predictions of the i_{th} class, the balanced accuracy can be defines as:

$$Balanced\ Acc = \frac{1}{C}\sum_{i=1}^{C}\frac{TP_i}{TP_i + FN_i} \tag{4.4}$$

In order to evaluate the performance of the neural network for each class, a confusion matrix of the true labels and the predictions is generated at the end of the test phase, that allows the values of true positives, true negatives, false positives and false negatives to be indirectly extracted.

Chapter 5

Experimental Work Results

This chapter begins with the analysis of the class distribution of the dataset, which provides insight into the balance and representation of different emotion labels. Following this, a detailed examination of the performance metrics calculated during the training, validation, and testing phases, including accuracy and balanced accuracy, was executed. These metrics offer a comprehensive evaluation of the model's effectiveness in recognizing emotions from facial images or their landmark representation. Additionally, confusion matrices are provided to evaluate the performance of the network to further illustrate the model's performance across different emotion classes.

5.1 Dataset Analysis

The class distribution analysis was performed for the training, validation and test sets, and calculated according to majority voting, the chosen methodology for selecting a single label for the images of a crowd-sourced label dataset.

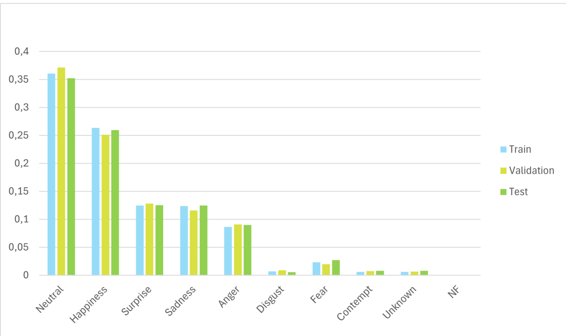


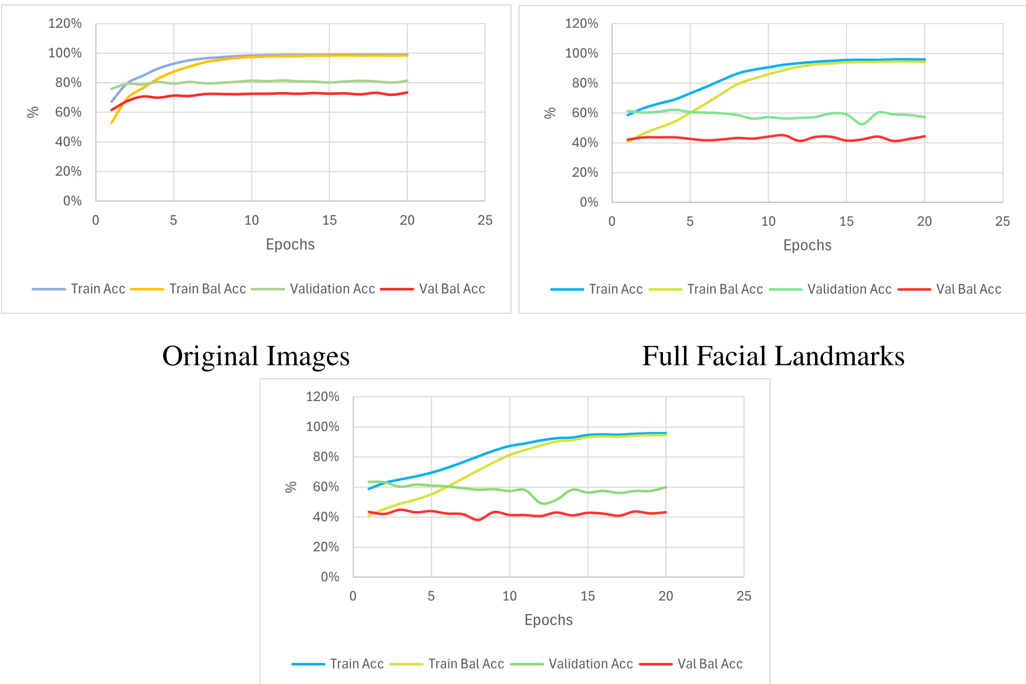
Figure 5.1: Dataset class distribution

From the analysis of Figure 5.1, Neutral is the class with the higher number of samples, being more than 35% in all the three sets. Following that is Happiness with over 25% and then Surprise and Sadness with a percentage of more than 10%. Disgust and Contempt are proven to be the least displayed emotion among the samples as they are less than 2,5% among the three sets of samples. The uncertainty among taggers achieves a percentage similar to Disgust and Contempt.

As for faces not found, those add up to an insignificant number of samples, with occurrences only on the train and validation sets. It is important to notice that the distribution of labels is somewhat equivalent through all the three sets and consequently will not be a cause for major differences in the results of the three phases.

5.2 Model Training

As stated before, the model's performance will be evaluated according to the values of train, validation and test accuracies and balanced accuracies. These metrics infer if the model is overfitting or underfitting and if it has the ability to generalize properly.
The values obtained for evaluation metrics of the initial sample images will serve as a baseline for comparison with the landmarked image's results.



Original Images

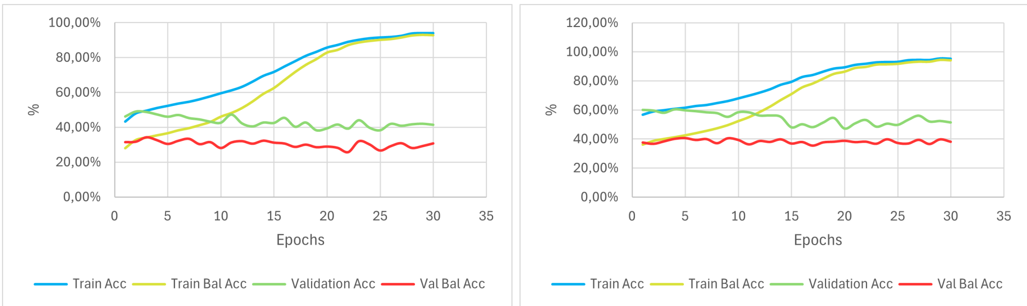
Full Facial Landmarks

Facial Features Landmarks

Figure 5.2: ResNet-18 training and validation results for FER+ dataset and its landmarked images

As can be seen in figure 5.2, the training accuracy and balanced accuracy reach almost 98% but the validation only about 81.35% and 73.29% for the original images, meaning the model is slightly overfitting. In terms of the number of epochs, it could be reduced to decrease the computational cost as the values begin to converge much sooner than 20 epochs.
The training phase with full facial landmarks follows the results of the original images, as both accuracy and balanced accuracy converge to values close to 100%, 95.85% and 94.42%. As for the validation phase, the metrics imply the performance of the model is worse an the values only reach 57.29% and 44.28%.

Evaluation the performance for the images comprised of the facial features landmarks and comparing with the full facial landmarks, the difference is almost insignificant. The train accuracy and balanced accuracy reach values of 95.92% and 94.54%, and the validation ones 59.84% and 43.20%, indicating that, during training, the model is able to correctly predict the labels better when it is only presented with the landmarks of the most relevant facial features for face emotion recognition instead of the 468 landmarks specified by MediaPipe. The results for the validation phase suggest the same, with a drawback that the model seem to be more biased.



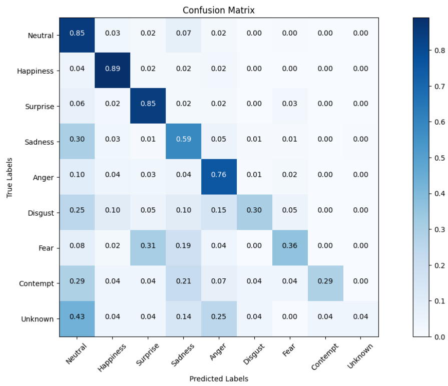
Eye Landmarks Mouth Landmarks

Figure 5.3: ResNet-18 training and validation results for eyes and mouth landmarks

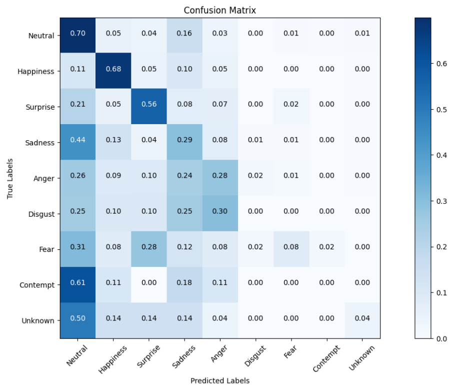
As for the use of only the landmarks of the eyes or mouth for face emotion recognition, the results in Figure 5.3 imply that the landmarks of the eyes are mouth are not sufficient for predicting facial expressions. The values of accuracy and balanced accuracy for the eye landmarks are 41.44% and 30.80% for the validation phase. As for the mouth landmarks, the results reach 51.46% and 38.14%. From these outcomes, one may conclude that the training with landmarks of the mouth enables the model to better predict emotions rather then the training with eye landmarks. However, both indicate a certain bias from the model as the value of balanced accuracy is lower in both instances.

As both accuracies for the validation phase are lower when compared to the ones of the training phase, this means that the model's is not as good at classifying unseen data as it is for the data with which it as trained.

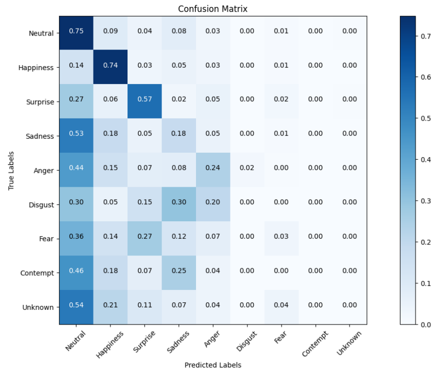
Regarding the information displayed by the confusion matrix of figure 5.4, it complies with the conclusion drawn from the values of the balanced accuracy for the train and validation phases. The model's overfitting and its performance is good for emotions such as Neutral, Happiness and Surprise for original images, full facial landmarks or facial features landmarks, but is only capable of recognizing Sadness and Anger for the original images and fails to correctly label Disgust, Fear and Contempt. This issue might be related to the class distribution of the dataset that includes more samples of the classes Neutral and Happiness. It is important to highlight that the values for the images with only the facial features landmarks regarding the classes Neutral, Happiness and Surprise are a bit higher when compared to the full facial landmarks images. The network is slightly biased towards the Neutral class, specially for full facial landmarks and facial features



Original Images



Full Facial Landmarks

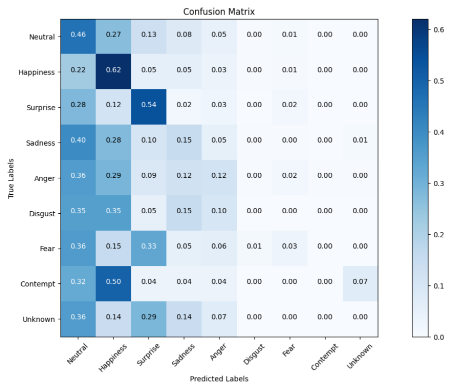


Facial Features Landmarks

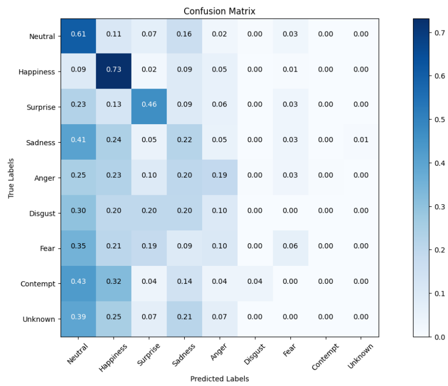
Figure 5.4: ResNet-18 confusion matrices for FER+ dataset and its landmarked images

landmarks, which could be caused by the higher number of samples labelled as Neutral when compared to the other labels.

Analysing the confusion matrices of Figure 5.5, the same conclusion from the accuracies can be drawn. The results indicate overfitting and a bias towards the class Neutral, although this class did not correspond to the best performance. One can conclude that the eyes fail to represent the characteristics of a Neutral face when compared to the mouth but shows average results for Happiness and Surprise. On the other hand, the mouth can represent a Neutral or emotion of Happiness and even Surprise, although the latter does not achieve results as good as the eyes. In both occasions, the model failed to correctly predict the labels of Sadness, Anger and specially Disgust, Fear and Contempt. This indicates that those emotions are not distinctly represented by the eyes or mouth or that the number of samples was not enough for the network to make a good prediction.



Eye Landmarks



Mouth Landmarks

Figure 5.5: ResNet-18 confusion matrices for eyes and mouth landmarks

Table 5.1: ResNet-18 test phase results of accuracy and balanced accuracy

Parameters	**Original**	**Full Facial**	**Facial Features**	**Eyes**	**Mouth**
Accuracy	79.21%	55.82%	57.10%	42.17%	50.67%
Balanced Accuracy	69.8%	42.74%	41.37%	32.16%	36.76%

Finally, with respect to the test phase, the values on 5.1 indicate a slight overfitting as they are somewhat lower than the validation phase results. Furthermore, the training with the original images remains the best in terms of both the evaluation metrics, followed sequentially by the images with full facial landmarks, facial features landmarks, mouth and ultimately eyes.

Chapter 6

Conclusions

This work serves the purpose of evaluating if an emotion manifested by a facial expression could be recognized solely on fiducial points and how the different facial features contribute to the expression of an emotion.

The previously exhibited results indicate a clear loss of performance for landmarked images, more so when there is a reduction on facial features, still achieving relatively good results for some classes. The network shows bias towards Neutral as it is the chosen prediction for the classes with a poor performance. It is the class with the best performance for all the different "datasets" except for the images with eye landmarks, where the class Happiness is the best. This indicates that the eyes are more capable of expressing cues for Happiness rather than the other emotions. A similar conclusion can be depicted for the mouth but in terms of Neutral and Happiness expressions.

The issues regarding the difference of performance for different labels could be explained either by the lack of a substantial amount of samples for classes other than Neutral and Happiness and by the inability of the landmarks to precisely mimic all the cues necessary for the correct prediction of an emotion. Furthermore, it is important to highlight that the relation between the different facial features is an important factor for face emotion recognition, as the accuracies corresponding to the facial features landmarks are better than the ones resulting from the training of only the eyes or mouth. Even if some facial features better represent certain emotion, a corrected prediction can be done with the contribution of all.

For future work, and to improve these results, a preprocessing step regarding data augmentation that performs a rotation or mirrors the image could be added to increase the number of samples and by making it harder for the model to memorize the training data, encouraging it to learn more robust features. Revisiting the labelling of the image samples could be another approach, as probabilistic label drawing has shown to achieve slightly better results for this dataset. Another option is resorting to a deeper neural network, that is composed of more layers, capable of a better performance but at the cost of higher computation power.

The work developed correlates to the goals defined by SDG, more specifically the sixteenth goal, as AFER improves interaction and inclusion of people with disabilities.

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